

EXPT NO:5	DEVELOP INTERACTIVE DASHBOARDS FOR DYNAMIC DATA EXPLORATION USING TABLEAU AND POWERBI
DATE: 24.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What is dynamic data exploration in BI tools, and why is it important?

Dynamic data exploration is the process of interacting with data using filters, slicers, drill-downs, and other features to analyze information from different perspectives. It is important because it helps users quickly identify trends, patterns, and insights, supporting better decision-making.

2. What are slicers and filters, and how do they enhance dashboard interactivity?

Slicers and filters allow users to display only the data they want to analyze. They improve dashboard interactivity by enabling users to focus on specific categories, dates, or regions without changing the original dataset, making analysis faster and easier.

3. How do Tableau and Power BI support drill-down and drill-through features?

Both Tableau and Power BI allow users to move from summary-level information to detailed data.

- **Drill-down** lets users explore hierarchical data, such as Year → Quarter → Month → Day.
- **Drill-through** (mainly in Power BI) allows users to navigate to another report page containing detailed information about a selected item. These features help users perform deeper analysis.

4. What is the role of parameters in creating interactive dashboards?

Parameters are user-defined controls that allow users to change values or switch between different measures and views dynamically. They make dashboards more flexible and interactive by allowing users to customize the analysis without modifying the underlying data.

5. Name some advanced interactivity features in Tableau and Power BI.

Some advanced interactivity features include:

- Filters and Slicers
- Parameters
- Drill-down
- Drill-through
- Tooltips
- Highlight Actions (Tableau)
- Bookmarks (Power BI)
- Interactive Dashboards

These features improve data exploration and help users gain deeper insights from the dashboard.

IN-LAB EXERCISE:

OBJECTIVE:

To design and develop interactive dashboards in Tableau and Power BI for dynamic data exploration, enabling users to analyze, filter, and drill down into data for deeper insights.

PROCEDURE:

1. Data Collection:

- Import datasets from sources such as Excel, CSV, SQL Server, or online APIs.
- Dataset: *Retail Sales Dataset* with fields such as Order Date, Region, Product Category, Sales, Profit, Quantity.

2. Data Preparation and Transformation:

- Ensure proper date formats for time-based analysis. ○
Handle missing or incorrect values.
- Create calculated fields (e.g., Profit Margin %, Sales Growth %, Year-to-Date Sales).

3. Data Aggregation and Hierarchies:

- Create hierarchies (e.g., Year → Quarter → Month → Day).
- Aggregate data by region, product category, or customer segment.

4. Designing Interactive Elements:

- **Filters & Slicers:** Allow filtering by product category, date range, or region. ○
Parameters: Let users switch between KPIs (e.g., Sales vs. Profit). ○ **Drill-down:** Enable users to explore data from summary to detail.
- **Tooltips:** Add contextual details when hovering over data points.

5. Dashboard Development:

- **Tableau:** Use interactive filters, parameter controls, and highlight actions.
- **Power BI:** Use slicers, bookmarks, and drill-through pages.
- Add KPI cards for key metrics like *Total Sales*, *Total Profit*, *Top Selling Product*. ○
Use maps for regional sales analysis, line charts for trends, and bar charts for category comparisons.

6. Testing Interactivity:

- Ensure slicers and filters update all relevant visuals dynamically.
- Test drill-down and drill-through functionality for accuracy.

7. Publishing and Sharing:

- **Tableau:** Publish on Tableau Public or Tableau Server.
- **Power BI:** Publish to Power BI Service and share via workspace access or public link.

Experiment Dataset:

Retail Sales Dataset

Fields Used:

- **Order Date** – Date of the order
- **Region** – Sales region
- **Product Category** – Category of product sold
- **Sales** – Sales revenue
- **Profit** – Net profit from the order
- **Quantity** – Units sold

OUTPUT :

Power BI :

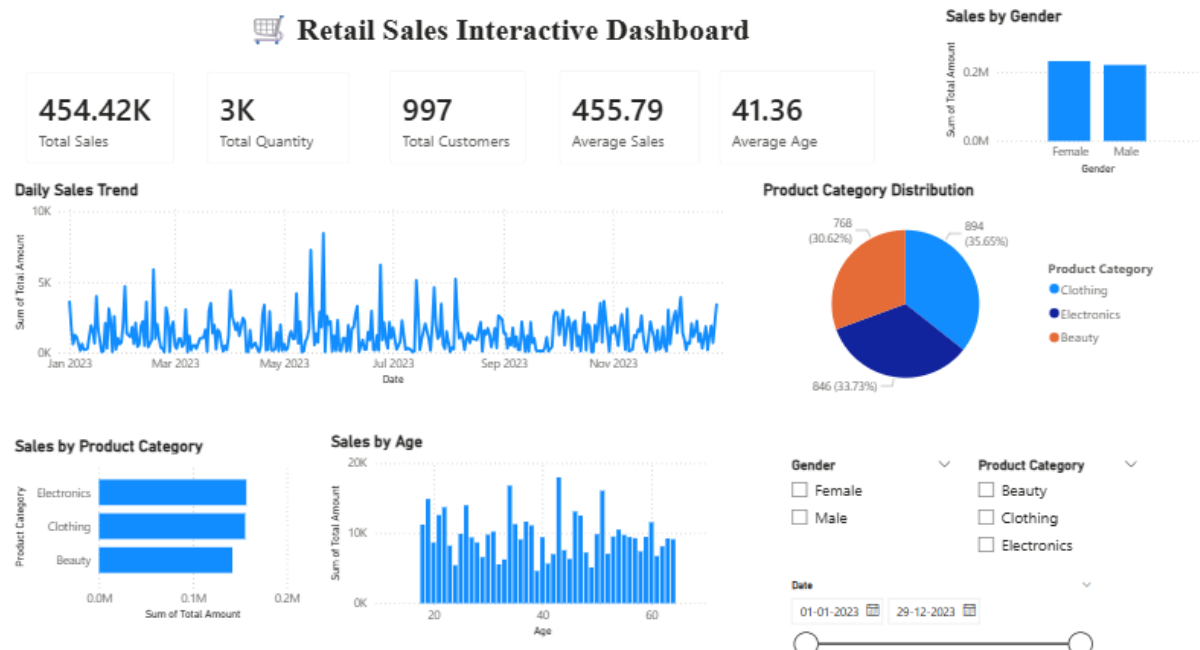
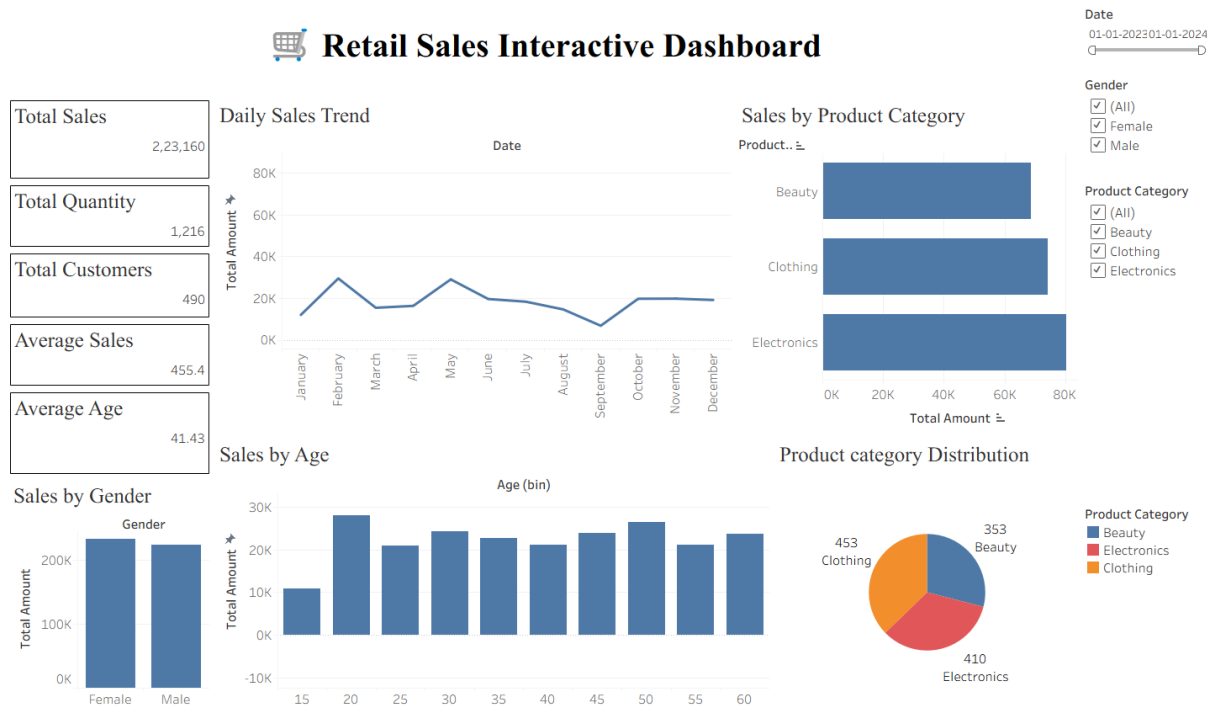


Tableau :



POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. How did interactivity improve the analysis experience in your dashboard?

Interactivity made the dashboard easy to explore. By using **date, gender, and product category filters**, I could instantly view sales for different selections without modifying the dataset. This made comparing sales trends and product performance much faster and more efficient.

2. Which interactive feature (filters, drill-down, parameters, etc.) was most useful and why?

The **filters** were the most useful feature. They allowed me to quickly analyze sales by product category, gender, and date, helping me compare different groups and identify trends without creating multiple reports.

3. What challenges did you face while implementing drill-down or dynamic filters?

One challenge was handling the **Age** field because Tableau initially treated it as a measure instead of a dimension. After converting it correctly, the chart displayed properly. I also ensured that the filters updated all worksheets so that the entire dashboard responded to user selections.

4. How do tooltips contribute to better dashboard storytelling?

Tooltips provide additional information when the user hovers over a chart. They display details such as **sales amount, product category, and quantity**, making the dashboard more informative without overcrowding the visuals.

5. What additional features would you add to make the dashboard more insightful?

To make the dashboard more useful, I would add:

- A Top 10 Products chart.
- A monthly sales growth percentage KPI.
- A customer age group analysis.
- A profit analysis and regional sales map if those fields were available in the dataset.

These features would provide deeper insights into sales performance and customer behaviour.

RESULT:

The retail sales dataset was successfully transformed, aggregated, and visualized into interactive dashboards in Tableau and Power BI. Filters, slicers, drill-downs, and parameters enabled dynamic exploration of sales and profit trends by region, product category, and time period. The resulting dashboards allowed for quick insights and deeper analysis, enhancing decision-making capabilities.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		